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EE Design 1 Technical Report

Power Supply Design

Introduction

In the Power Supply Design module, we are tasked with creating two voltage regulators, one utilizing an LM7805 linear regulator, and the second using an LM2576 adjustable switching regulator. Our goal is to create designs for both to output 5V with 5mA & 100mA output currents as options, and to record their characteristics in both simulation and practical application to determine the ideal use-case for either chip.

Methods

Starting our design, we start by designing the circuit in LTSpice to ensure correct operation and output. To model our circuit in LTSpice, we must begin by modeling our RIGOL 2702 power supply, which is done using an ideal voltage source to model our 120VAC output from the wall receptacle. This then goes to a 130:1 transformer, due to the relationship below:

$$\frac{V_{primary}}{V_{secondary}} = \sqrt{\frac{L_1}{L_2}}$$

We find that a ratio of 130:1 would give us a voltage output ratio from primary to secondary of 11.4:1, which means that our output is $120V/11.4V \cong 10.52V$, which when accounting for our full-bridge rectification forcing current through 2 diodes, which drops the voltage by 0.7V per diode, gives us $10.52V - 1.4V = 9.12VAC$.

This is followed by a bypass capacitor, which removes high-frequency noise from our input pin by shorting it to ground, and supplements the charge on the VIN pin to allow the IC to convert AC to DC. A 1000uF capacitor is a solid middle-ground of medium-large capacitors that are large enough to store significant charge without causing intense in-rush current upon activation.

On the output side for the LM2576 adjustable, we provide a Zener diode to utilize non-linear reverse breakdown, effectively providing a hard limit to the output voltage, releasing the excess to ground (TI 2026.) The inductor is important for a buck converter, to store energy and allow for voltage smoothing (Gemini/GPT). After this, we provide another bypass capacitor for noise

reduction. The adjustable version of the LM2576 also requires in section 8.2 of the datasheet to provide two resistors in series, with the top node connecting to output, the middle to feedback, and the bottom to ground. Feedback forces a reference voltage of 1.23, and the output provides a voltage of:

$$V_{out} = V_{ref} * \left(1 + \frac{R2}{R1}\right)$$

Schematics are shown below for the LM7805 (linear regulator), and LM2576 (switching regulator.)

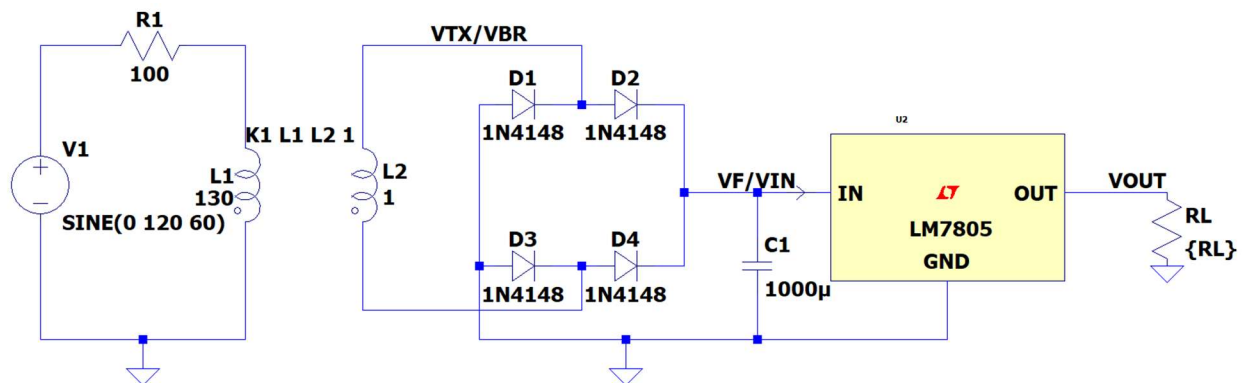


Figure 1: Linear Regulator LTSpice Schematic

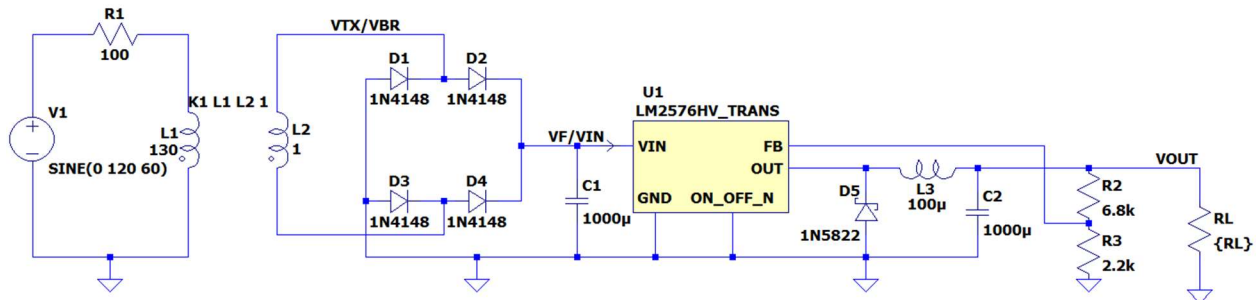


Figure 2: Switching Regulator LTSpice Schematic

The linear regulator

Results

Linear Regulator Simulated			Switching Regulator Simulated		
	47 Ω	1000 Ω		47 Ω	1000 Ω
V_xfrmr (mV)	-20.186	-20.186	V_xfrmr (mV)	-20.186	-20.186
V_bridge (V)	4.4275	4.4275	V_bridge (V)	4.4275	4.4275
V_filter (V)	9.0519	9.0519	V_filter (V)	9.0519	9.0519
V_in (V)	7.6949	8.7545	V_in (V)	7.9797	8.906
V_out (V)	5.0028	5.0028	V_out (V)	5.0186	5.0538
I_in (mA)	111.43	10.02	I_in (mA)	76.053	3.5881
I_out (mA)	106.44	5.0028	I_out (mA)	106.78	5.0538
Linear Regulator Practical			Switching Regulator Practical		
	47 Ω	1000 Ω		47 Ω	1000 Ω
V_in (V)	8.4909	8.4967	V_in (V)	8.6324	7.914
V_out (V)	5.1448	5.2484	V_out (V)	4.9216	5.1795
I_in (mA)	110.285	10.3528	I_in (mA)	90.019	9.8578
I_out (mA)	102.897	5.4087	I_out (mA)	105.346	5.2099
P_in (W)	0.936419	0.087965	P_in (W)	0.77708	0.078015
P_out (W)	0.529384	0.028387	P_out (W)	0.518471	0.026985
Efficiency	56.53%	32.27%	Efficiency	66.72%	34.59%
Noise (mVpp)	27.2	21.6	Noise (mVpp)	824	84.8

Figure 3: Simulation & Practical Data Measurements

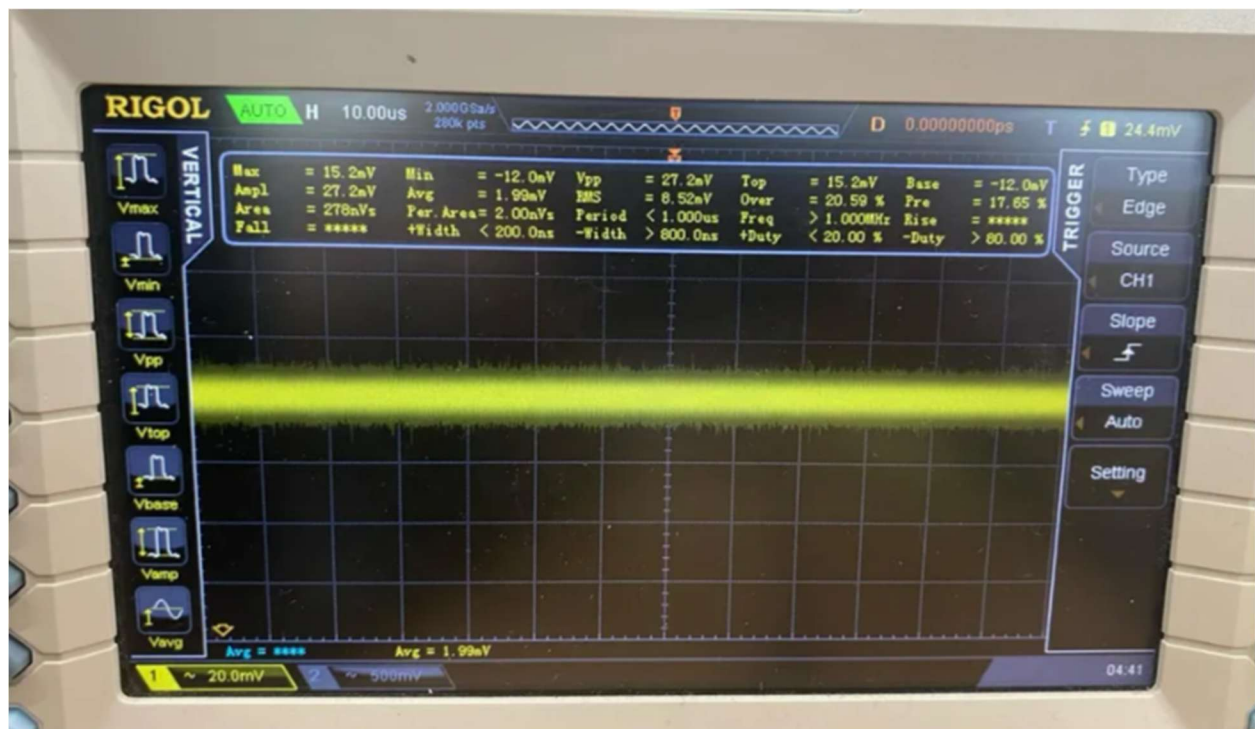


Figure 4: 47 Ω Linear Regulator AC Coupling Oscilloscope

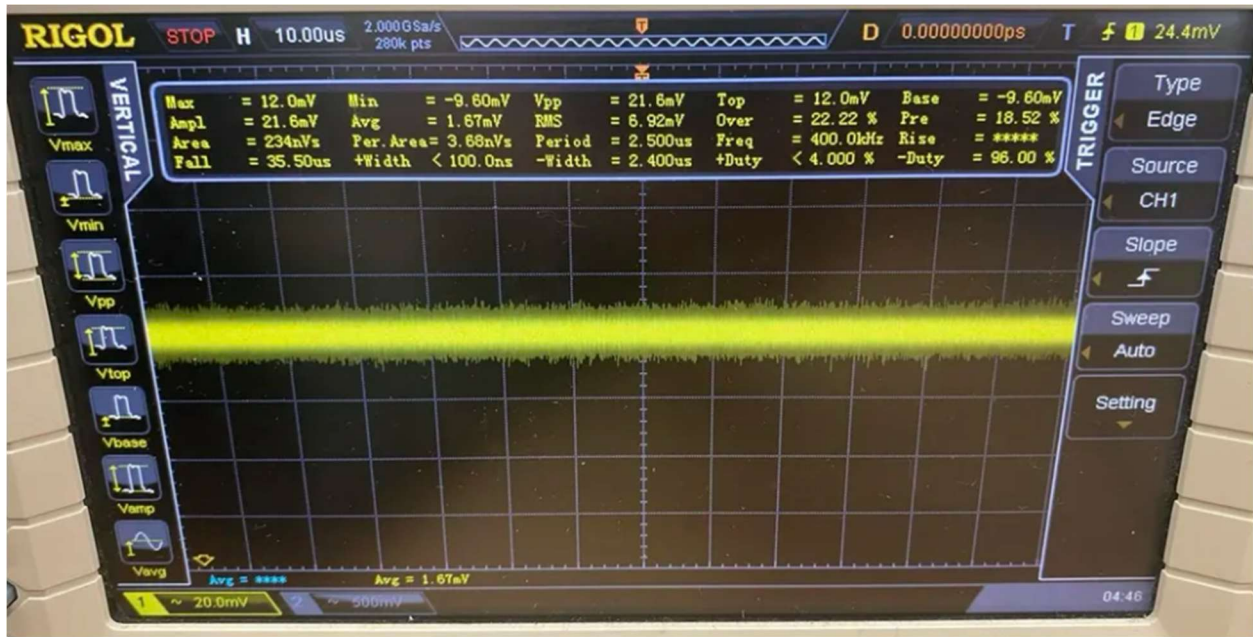


Figure 5: 1k Ω Linear Regulator AC Coupling Oscilloscope

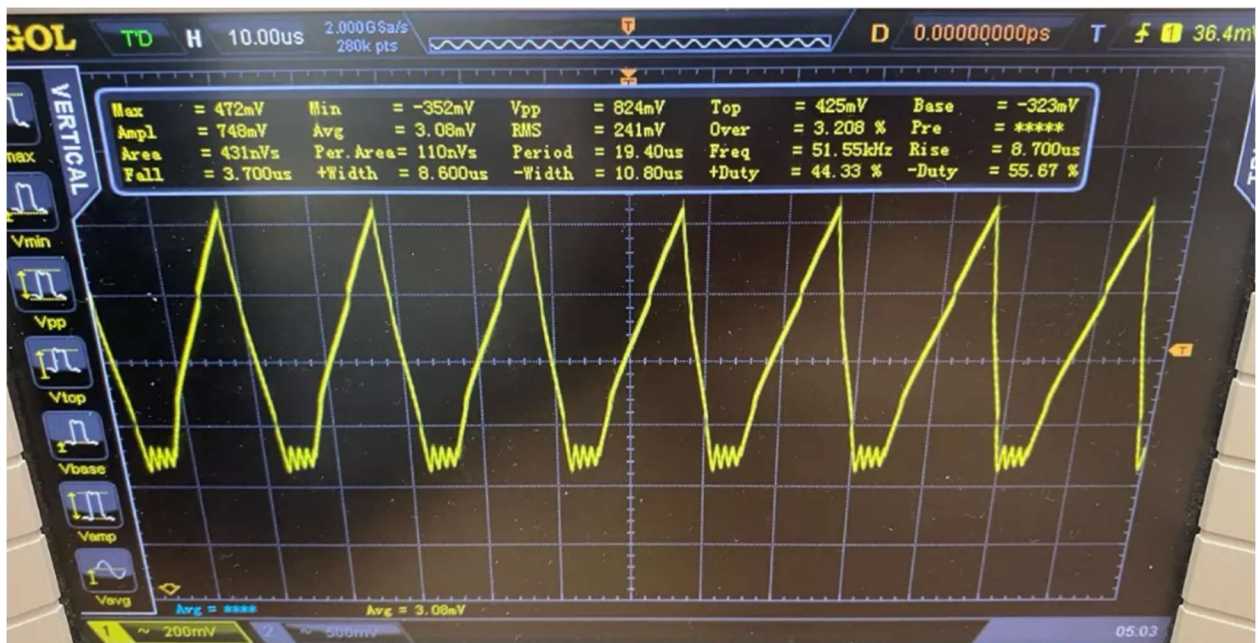


Figure 6: 47 Ω Switching Regulator AC Coupling Oscilloscope



Figure 7: 1k Ω Switching Regulator AC Coupling Oscilloscope

From our data, we find that there are some discrepancies, first we find that the voltage outputs are not the ideal output of the chip's internals. This is likely due to non-idealities of the chip as it is exposed to heat and internal references are slightly altered, and potential inductance interference from the length of wires connecting components (Gemini/GPT.)

We also read that we can have significant noise, as mentioned above this is contributed to by the inductance of our wires, as tested when using shorter wires we experience less noise comparatively to the longer wires.

Per our results, we can see that both regulators are within spec for voltage and possible slightly without of range of spec for the current, this is likely due to the resistor being 47 ohm rather than 50 ohm. Particularly, both are better with the lower loads, with the switching regulator being better for higher loads and the linear typically being better for lower loads (due to a fixed input current requirement for the switching vs a portion of the current being drawn/burnt.) (TI 2014), it is likely that there is some discrepancy due to wires and non-idealities of the practical assembly.

The switching regulator is much noisier, with up to over 30 times the noise Vpp, this is due to the switching operation versus a pass transistor in the linear, causing less noise introduction (TI 2014.) Noise scales inversely with load, leading to the 47 Ω load switching regulator to be the noisiest.

Conclusion

In this design assignment, we find that linear regulators are very good at low loads but drop off quickly with higher loads. The switching regulator performed similarly, but likely had assembly error. The switching regulator has much more severe noise due to active switching of internal transistors. To design these circuits, we learnt about Zener diodes non-linearity to prevent electrostatic discharge, and the use-case of capacitors and inductors in regulating circuitry, along with full-bridge rectifiers to convert AC to DC.

This assignment reinforced the emphasis on regulators being an integral part of power architecture of electronics, and introduced us to their applications and how to simulate them from the source power down-stream, emphasizing our usage of LTSpice and RIGOL electronics oscilloscopes, power supplies, waveform generators, and miscellaneous tools.

References

1. Texas Instruments. (2014). *Linear versus Switching Regulators in Industrial Applications with a 24-V Bus* (Application Report SLYT527). Texas Instruments. <https://www.ti.com/lit/pdf/slyt527>
2. Texas Instruments. (2026). *Zener Diodes Introduction* (Document SLVAG00). Texas Instruments. <https://www.ti.com/lit/SLVAG00>
3. Google Gemini and ChatGPT referenced above.